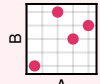


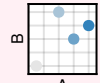


AlgebraOfGraphics.jl Cheat Sheet

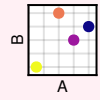
data(df_one) * mapping(:A, :B)



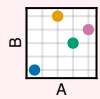
* visual(Scatter)



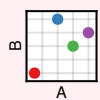
* mapping(color=:C)
* visual(Scatter)



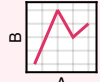
* mapping(color=:C)
* visual(Scatter) |>
draw(scales(Color=(:colormap=:plasma)))



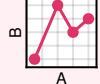
* mapping(color=:D)
* visual(Scatter)



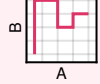
* mapping(color=:D)
* visual(Scatter) |>
draw(scales(Color=(:palette=:Set1_5)))



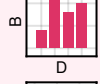
* visual(Lines)



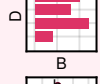
* visual(ScatterLines)



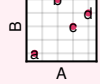
* visual(Stairs)



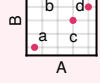
* visual(BarPlot)



* visual(BarPlot,direction=:x)

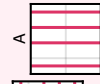


* mapping(text=:D=>verbatim)
* visual(Makie.Text)



* mapping(text=:D=>verbatim)
* visual(Annotation)

data(df_one) * mapping(:A)

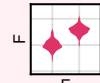


* visual(HLines)

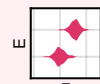


* visual(VLines)

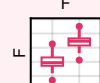
data(df_two) * mapping(:E, :F)



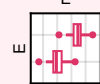
* visual(Violin)



* visual(Violin,orientation=:horizontal)

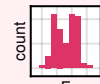


* visual(BoxPlot)

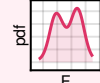


* visual(BoxPlot,orientation=:horizontal)

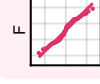
data(df_two) * mapping(:F)



* histogram()

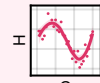


* AoG.density()



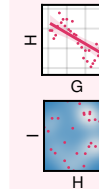
* visual(QQNorm)

data(df_three) * mapping(:G, :H)



* smooth()

```
df_one = (A=[1,4,6,8], B=[2,6,4,5], C=[3,2,1,0], D=["a","b","c","d"])
df_two = (E=repeat(["e","f"],inner=50), F=[randn(50);randn(50)].+3)
df_three = (G=1:30, H=sin.(range(0,2pi,30)).+rand.(), I=cos.(range(0,2pi,30)).+rand.())
df_four = (J=repeat(1:3,3), K=repeat(1:3,inner=3), L=[0,1,2,0.5,2,4,1,4,5])
df_five = (M=repeat(1:3,3), N=[0,2,3,0.5,3.5,5,1,5,7], O=repeat(["g","h","i"],inner=3))
df_six = (P=1:4, Q=["A","B","A","B"], R=-1:1, S=[0.5,0.6,0.3,0.7], T=[5.1,3.9,2.7,1.5], U=[1,1,2,2])
```



* linear()



* AoG.density()

data(df_four) * mapping(:J, :K, :L)



* visual(Heatmap)



* contours(bands=4)



* filled_contours(bands=4)

data(df_five) * mapping(:M, :N)



* mapping(group=:O)
* visual(Lines)



* mapping(color=:O)
* visual(Lines)



* mapping(linestyle=:O)
* visual(Lines)



* mapping(marker=:O)
* visual(ScatterLines)



* mapping(color=:O,dodge=:O)
* visual(BarPlot)



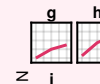
* mapping(color=:O,stack=:O)
* visual(BarPlot)



* mapping(row=:O)
* visual(Lines)

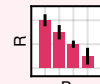


* mapping(col=:O)
* visual(Lines)

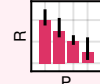


* mapping(layout=:O)
* visual(Lines)

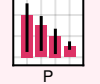
data(df_six) * mapping(:P, :Q, :R, :S, :T, :U)



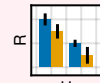
* (mapping(:P,:R) * visual(BarPlot)
+ mapping(:P,:R,:S) *
visual(Errorbars))



* (mapping(:P,:R) * visual(BarPlot)
+ mapping(:P,:R,:S,:S=>x->2x) *
visual(Errorbars))

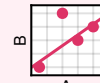


* mapping(group=:O)
* visual(Lines)

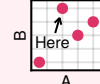


* (mapping(:U,:R,dodge=:Q,color=:Q) *
visual(BarPlot)
+ mapping(:U,:R,:S,dodge_x=:Q) *
visual(Errorbars))

Others



data(df_one) * mapping(:A,:B) *
visual(Scatter) + mapping(2,0.4) *
visual(ABLines)



mapping(-5,-17,4,6) * visual(Annotation,
text="Here", style=Ann.Styles.LineArrow())